AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Administration

FIN342
Financial & Securities Markets
Version C

Mid-term Examination

2024-2025 / Spring

5-April-2025

Number of Exam Pages: 4 (including this cover sheet)	Time Allowed: 2 hours

Instructions:

1- Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
2- Mobile phones must be in "SWITCHED **OFF**" mode throughout the examination.
3- This is a **CLOSED** book examination. NO documents of any kind are allowed.
4- You **MUST** answer the questions using only the answer booklets provided
5- You **MUST** return the examination paper together with your answer booklets.
6- Do **NOT** write your answers in pencil except graphs/sketches
7- Students are **NOT** allowed to use any type of dictionary or mobile phones during the examinatio
8- Students are allowed to use ***non-programmable*** calculators only (if needed).
9- Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.

CamScanner

Solve All questions

Question 1: (7 marks) (5 min)

Explain the structure of financial markets, focusing on primary and secondary markets. (Provide example for each market).

Question 2: (12 marks) (30 min)

Consider a bond with an 11% annual coupon and a face value of $1,000. Complete the following table:

Years to Maturity	Discount Rate	Current Price
3	10	?
3	11	?
6	15	?
9	14	?
10	13	?

What is the relationship between yield to maturity and the current market value?

Question 3: (10 marks) (30 min)

You are considering purchasing either a 5-year corporate bond or a 5-year municipal bond. The corporate bond has a 7% annual coupon rate, a par value of $1,000, and is currently yielding 8.5%. The municipal bond has an 8% annual coupon rate, a par value of $1,000, and is currently yielding 9%.

Assume that your marginal tax rate is 30%. Which bond would be more beneficial for you to purchase?

Question 4: (14 marks) (30 min)

You own a $1,500-par zero-coupon bond with 6 years remaining until maturity. You plan to sell the bond in one year. The required yield next year is expected to follow this probability distribution:

Probability	Required Yield
0.25	6.50%
0.15	6.75%
0.20	7.00%
0.25	7.25%
0.15	7.50%

a. What is your expected price when you sell the bond?
b. What is the standard deviation?

CamScanner

Question 5: (7 marks) (15 min)

What would be your annualized discount rate % and your annualized investment rate % on the purchase of a 182-day Treasury bill for $3,000 that pays $3,950 at maturity?

End of Questions

OU EGYPT

CamScanner

Formulas

Coupon Payment=

$$P = \frac{C}{(1+i)} + \frac{C}{(1+i)^2} + \frac{C}{(1+i)^3} + \cdots + \frac{C}{(1+i)^n} + \frac{F}{(1+i)^n}$$

Expected return= Sum(Prob × Price)

Variance= **Sum (Prob × (Price – Exp. Price)^2)**

Standard Deviation=$\sqrt{Prob(price - exp.price)^2}$

Liquidity Premium theory for n period bond=

$$i_{nt} = \frac{i_t + i^e_{t+1} + i^e_{t+2} + \cdots + i^e_{t+(n-1)}}{n} + l_{nt}$$

Treasury Bills discount rate=

$$i_{discount} = \frac{F - P}{F} \times \frac{360}{n}$$

Treasury bill investment rate=

$$i_{yt} = \frac{F - P}{P} \times \frac{365}{n}$$

EGYPT

CamScanner